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FIXTURE AND METHOD FOR DETERMINING POSITION OF A TARGET IN A REACTION CHAMBER

Abstract

A fixture includes a frame, a leveling plate, a bracket, and a laser profiler. The frame is arranged for fixation above a reaction chamber arranged to deposit a film onto a substrate. The leveling plate is supported on the frame. The bracket is supported on the leveling plate. The laser profiler is suspended from the bracket, overlays the reaction chamber, and has a field of view that extends through the leveling plate and the frame to determine position of a target within the reaction chamber. Semiconductor processing systems and methods of determining position of targets within reaction chambers in semiconductor processing systems are also described.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a divisional of, and claims priority to and the benefit of, U.S. patent application Ser. No. 17/549,311, filed Dec. 13, 2021 and entitled “FIXTURE AND METHOD FOR DETERMINING POSITION OF A TARGET IN A REACTION CHAMBER,” which is a nonprovisional of, and claims priority to and the benefit of, U.S. Provisional Patent Application No. 63/126,176, filed Dec. 16, 2020 and entitled “FIXTURE AND METHOD FOR DETERMINING POSITION OF A TARGET IN A REACTION CHAMBER,” all of which are hereby incorporated by reference herein.

FIELD OF INVENTION
[0002] The present disclosure generally relates to semiconductor processing systems, and more particularly, to determining position of targets within reaction chambers in semiconductor processing systems.
BACKGROUND OF THE DISCLOSURE
[0003] Semiconductor processing systems are commonly used to deposit films onto substrates, such as during the fabrication of very large-scale integrated circuits or power electronics. Film deposition in such systems is generally accomplished by supporting a substrate within a reaction chamber and flowing a precursor gas through the reaction chamber to deposit a film onto the substrate. The reaction chamber typically flows the precursor gas through the reaction chamber according to a predetermined gas flow pattern during deposition of the film onto the substrate. The reaction chamber also typically maintains predetermined thermal conditions within the reaction chamber environment during deposition of the film onto the substrate.
[0004] In some film deposition techniques deviation from the predetermined gas flow pattern and/or the predetermined thermal conditions may introduce properties and characteristics in the film that differ from those intended, such as thickness variation and/or variation in the electrical properties of the deposited film. The gas flow pattern and/or thermal conditions within a particular the reaction chamber may vary, for example, according to deviation in the ‘as built’ positioning of certain structures within the reaction chamber in comparison to the intended position(s) of the structures within the reaction chamber. The gas flow pattern and/or thermals conditions within a reaction chamber may also vary according to deviation in the positioning of the substrate within the reaction chamber, e.g., shifts and/or decentering of the placement of the substrate on the substrate holder in advance of the deposition process. For that reason, the ‘as built’ performance within reaction chambers may be compared against certain process limits, and adjustments made to the reaction chamber, prior to the reaction chamber being qualified for production.
[0005] Such systems and have generally been satisfactory for their intended purpose. However, there remains a need in the art for improved fixtures, semiconductor processing systems, and methods of determining placement of targets within reaction chambers in semiconductor processing systems. The present disclosure provides a solution to this need.
SUMMARY OF THE DISCLOSURE
[0006] A fixture is provided. The fixture includes a frame, a leveling plate, a bracket, and a laser profiler. The frame is configured for fixation above a reaction chamber that is configured to deposit a film onto a substrate. The leveling plate is supported on the frame. The bracket is supported on the leveling plate. The laser profiler is suspended from the bracket, overlays the reaction chamber, and has a field of view extending through the leveling plate and the frame to determine position of a target within the reaction chamber.

[0007] In certain examples, the frame may include a first stringer, a second stringer offset from the first stringer, and two or more joists extending between the first stringer and the second stringer. The first stringer and the second stringer may longitudinally span the reaction chamber.
[0008] In certain examples, the bracket may include a U-shaped body having a lower tine connected to an upper tine by a web. The upper tine may have a profiler seat extending along the upper tine, the lower tine may have a bracket seat extending along the lower tine, and the bracket seat may extend in parallel with the profiler seat.
[0009] In certain examples, the bracket seat may abut the leveling plate and the laser profiler may be suspended from the profiler seat and above the lower tine of the U-shaped body.
[0010] In certain examples, the bracket may include an end cap. The end cap may be fixed to ends of the upper tine and the lower tine opposite the web. The end cap may have a cable passthrough. A cable may extend through the cable passthrough and be electrically connected to the laser profiler to communicate height measurements acquired by the laser profiler.
[0011] In certain examples, the fixture may include a first leveling screw and one or more second leveling screw. The first leveling screw may be threadedly received in the leveling plate and space the leveling plate apart from the frame. The one or more second leveling screw may be threadedly received in the leveling plate and space the leveling plate apart from the frame. The one or more second leveling screw may be offset from the first leveling plate on a mounting surface of the leveling plate. In certain examples the fixture may include at least three leveling screws.
[0012] In certain examples, the fixture may include a first level and a second level. The first level may extend longitudinally along a mounting surface of the leveling plate. The second level may extend laterally along the mounting surface of the leveling plate. The second level may be orthogonal relative to the first level. The first level may be fixed relative to the level plate. The second level may be fixed relative to the leveling plate.
[0013] In certain examples, the laser profiler may include a laser source, an expansion lens, collection optics, and a sensor. The expansion lens may be optically coupled to the laser source and configured to widen a beam incident upon the expansion lens into a line and project the line onto the target. The collection optics may be optically coupled to the expansion lens by the target. The sensor may be optically coupled to the collection optics and configured to determine height of the laser profiler from the target along the line.
[0014] In certain examples, the laser profiler source may include a (a) a visible laser source, (b) a blue wavelength laser source, or (c) a 405-nanometer laser source.
[0015] In certain examples, the bracket may have a bracket aperture extending therethrough, the expansion lens may be aligned between the laser source and the bracket aperture to project the line onto the target, and the collection optics may be aligned between the bracket aperture and the sensor to collect and communicate illumination reflected from the target onto the sensor.
[0016] In certain examples, the leveling plate may have one or more of a shelf mount, a one-piece ring mount, a susceptor mount, a wafer center mount, and a wafer periphery mount. The shelf mount may have a shelf mount aperture and may be configured to seat the bracket. The one-piece ring mount may be longitudinally and/or laterally offset from the shelf mount, may have a one-piece ring mount aperture, and may be configured to seat the bracket. The susceptor mount may be longitudinally and/or laterally offset from the one-piece ring mount, may have a susceptor mount aperture, and may additionally be configured to seat the bracket. The wafer center mount may be longitudinally and/or laterally offset from the susceptor mount, may have a wafer center aperture, and may also be configured to seat the bracket. The wafer periphery mount may be radially offset from the wafer center mount, may have a wafer periphery mount aperture, and may be further configured to seat the bracket.
[0017] A semiconductor processing system is provided. The semiconductor processing system includes a reaction chamber, a shelf, a one-piece ring, susceptor, and a fixture as described above. The shelf is fixed within an interior of the reaction chamber. The one-piece ring is positioned

within the reaction chamber, abuts the shelf, and has a one-piece ring aperture extending therethrough. The susceptor is supported within the interior of the reaction chamber for rotation about a rotation axis that extends through the one-piece ring aperture, The frame of the fixture supports the leveling plate above the reaction chamber such that at least one of the shelf, the one-piece ring, the susceptor, or a wafer positioned between the susceptor and the leveling plate is in the field of view of the laser profiler.

[0018] A method of determining position of a target within a reaction chamber of a semiconductor processing system is provided. The method includes, at a fixture as described above, leveling the leveling plate above the reaction chamber relative to gravity, projecting a line into the reaction chamber and onto a target within the reaction chamber, and acquiring height of the laser profiler above the target at a first point and a second point along the line. The second point spaced apart from the first point along the line and position of the target within the reaction chamber is determined using height of the laser profiler above the target at the first point and the second point along the line.

[0019] In certain examples, the target may include a shelf fixed within an interior of the reaction chamber. Determining position of the target may include determining at least one of height and level of the shelf relative to the leveling plate.

[0020] In certain examples, the target may include a one-piece ring seated within the reaction chamber. Determining position of the target may include determining at least one of height and level of the one-piece ring relative to the leveling plate.

[0021] In certain examples, the target may include a susceptor supported within the reaction chamber for rotation about a rotation axis. Determining position of the target may include determining at least one of (a) height of the susceptor relative to the leveling plate, (b) level of the susceptor relative to the leveling plate, (c) centering of the susceptor relative to the one-piece ring (d) wobble of the susceptor during rotation about the rotation axis, and (e) runout of the susceptor during rotation about the rotation axis.

[0022] In certain examples, the method may include supporting a wafer on a susceptor supported within the reaction chamber or for rotation about a rotation axis and displacing the wafer from the susceptor using a plurality of lift pins translatable along the rotation axis relative to the susceptor. Projecting the line onto the target may include projecting the line onto the wafer and determining position of the target may further include identifying contact height of the plurality lift pins with the wafer prior to displacement of the wafer from the susceptor.

[0023] In certain examples, projecting the line onto the target may include projecting the line onto a center portion of the wafer and the method may further include tuning speed of movement of the plurality of lift between an extended position and a retracted position along the rotation axis during loading of wafers onto the susceptor.

[0024] In certain examples, projecting the line onto the target may include projecting the line onto a peripheral portion of the wafer and the method may further include determining level of the plurality of lift pins relative to the leveling plate using the contact height of the plurality of lift pins.

[0025] In certain examples, the contact height may be a first contact height and the method may include acquiring a second contact height of the plurality of lift pins using the laser profiler, and level of the plurality of lift pins may be determined using both the first contact height and the second contact height of plurality of lift pins.

[0026] This summary is provided to introduce a selection of concepts in a simplified form. These concepts are described in further detail in the detailed description of example embodiments of the disclosure below. This summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter.

Description

BRIEF DESCRIPTION OF THE DRAWING FIGURES

[0027] These and other features, aspects, and advantages of the present disclosure are described below with reference to the drawings of certain embodiments, which are intended to illustrate and not to limit the invention.

[0028] FIG. **1** is a cross-sectional side view of a semiconductor processor system with a reaction chamber and a fixture constructed in accordance with the present disclosure, schematically showing the fixture exploded away from the reaction chamber;

[0029] FIG. **2** is a cross-sectional end view of the fixture and the reaction chamber of FIG. **1**, schematically showing a laser profiler suspended from a bracket and supported on a leveling plate and a frame such that a target within the reaction chamber is in a field of view of the laser profiler;

[0030] FIG. **3** is a plan view of the reaction chamber and a portion of the fixture of FIG. **1**, schematically showing the frame spanning the reaction chamber and supported at longitudinally opposite ends on an injection flange and an exhaust flange of the reaction chamber;

[0031] FIG. **4** is a cross-sectional side view of a portion of the fixture of FIG. **1** including the bracket, showing a U-shaped body and an end cap of the bracket;

[0032] FIG. **5** is a schematic view of the laser profiler of FIG. **1**, showing the laser profiler projecting a line into the reaction chamber to determine position of the target;

[0033] FIG. **6** is a plan view of the reaction chamber and a portion of the fixture of FIG. **1** including the leveling plate, schematically showing a shelf mount located on the leveling plate and overlaying a shelf positioned within the reaction chamber according to an example;

[0034] FIG. **7** is a plan view of the reaction chamber and a portion of the fixture of FIG. **1** including the leveling plate, schematically showing a one-piece ring mount located on the leveling plate and overlaying a one-piece ring positioned within the reaction chamber;

[0035] FIG. **8** is a plan view of the reaction chamber and a portion of the fixture of FIG. **1** including the leveling plate, schematically showing a susceptor located on the leveling plate and overlaying a susceptor positioned within the reaction chamber;

[0036] FIG. **9** is a plan view of the reaction chamber and a portion of the fixture of FIG. **1** including the leveling plate, schematically showing a wafer center mount located on the leveling plate and overlaying a wafer positioned within the reaction chamber;

[0037] FIGS. **10** and **11** are side views of the reaction chamber and fixture of FIG. **9**, schematically showing the laser profiler acquiring height measurements of the laser profiler above the wafer to determine contact height of lift pins positioned within the reaction chamber;

[0038] FIG. **12** is a plan view of the reaction chamber and a portion of the fixture of FIG. **1** including the leveling plate, schematically showing a wafer periphery mount located on the leveling plate and overlaying a wafer positioned within the reaction chamber;

[0039] FIGS. **13** and **14** are side views of the reaction chamber and fixture according to the example of FIG. **12**, showing the laser profiler acquiring height measurements of the wafer to determine to determine level of lift pins positioned within the reaction chamber; and

[0040] FIG. **15** is a block diagram of a method of determining position of a target within a reaction chamber of a semiconductor processing system, showing operations of the method.

[0041] It will be appreciated that elements in the figures are illustrated for simplicity and clarity and have not necessarily been drawn to scale. For example, the dimensions of some of the elements in the figures may be exaggerated relative to other elements to help improve understanding of illustrated embodiments of the present disclosure.

DETAILED DESCRIPTION OF EXEMPLARY EMBODIMENTS

[0042] Reference will now be made to the drawings wherein like reference numerals identify similar structural features or aspects of the subject disclosure. For purposes of explanation and illustration, and not limitation, a partial view of an example of a fixture in accordance with the disclosure is shown in FIG. **1** and is designated generally by reference character **100**. Other examples of fixtures, semiconductor processing systems, and methods of determining position of targets within reaction chambers in semiconductor processing systems in accordance with the present disclosure, or aspects thereof, are provided in FIGS. **2-15**, as will be described. The fixtures, semiconductor processing systems, and methods of determining position of targets within reaction chambers in semiconductor processing systems described herein may be used to control properties of films deposited onto substrate in such systems, such as in films deposited onto wafers using epitaxial deposition techniques in atmospheric chemical vapor deposition (CVD) semiconductor processing systems, though the present disclosure is not limited to CVD semiconductor processing system or epitaxial deposition techniques in general.

[0043] Referring to FIG. **1**, a semiconductor processing system **10** is shown. The semiconductor processing system includes a reaction chamber **12**, an injection flange **14**, an exhaust flange **16**, a first precursor source **18**, and a second precursor source **20**. The semiconductor processing system **10** also includes a purge source **22**, a shelf **24**, a one-piece ring **26**, and a susceptor **28**. The semiconductor processing system **10** additionally includes a spider **30**, a shaft **32**, a drive module **34**, and a plurality of lift pins **36** (shown in FIG. **3**). Although a particular type of semiconductor processing system is shown in FIG. **1** and referred to throughout the present disclosure, e.g., an atmospheric CVD semiconductor processing system employed for epitaxial thin film deposition techniques, it is to be understood and appreciated that other types of semiconductor processing systems may also benefit from the present disclosure, such as atomic layer deposition systems by way of non-limiting example.

[0044] The reaction chamber **12** is configured to deposit a film **62** onto a substrate **64** and has an injection end **38**, an exhaust end **40**, and an interior **42**. The exhaust flange **16** is connected to the exhaust end **40** of the reaction chamber **12** and fluidly couples the interior **42** of the reactor to the environment external to the reaction chamber **12**. The injection flange **14** is connected to the injection end **38** of the reaction chamber **12** and is fluidly coupled to the exhaust flange **16** by the reaction chamber **12**. The injection flange **14** fluidly couples the first precursor source **18**, the second precursor source **20**, and the purge source **22** to the reaction chamber **12**. In certain examples, the reaction chamber **12** may formed from quartz. In accordance with certain examples, the reaction chamber **12** may have one or more ribs protruding from the reaction chamber **12** and extending laterally about the reaction chamber **12**. The reaction chamber **12** may be as shown and described in U.S. Pat. No. 7,108,753 to Wood, issued on Sep. 19, 2006, the contents of which is incorporated herein by reference in its entirety.

[0045] The first precursor source **18** is connected to the injection flange **14** and is configured to communicate a first precursor **44** to the reaction chamber **12**, e.g., a silicon-containing precursor, to the reaction chamber **12**. The second precursor source **20** is connected to the injection flange **14** and is configured to communicate a second precursor **46**, e.g., a dopant-containing precursor, to the reaction chamber **12**. The purge source **22** is connected to the injection flange **14** and is configured to communicate a purge gas **48**, e.g., hydrogen gas, to the reaction chamber **12**. It is contemplated that at least one of the first precursor **44** and the second precursor **46** be selected to deposit a film **62** onto a wafer **64** supported within the interior **42** of the reaction chamber **12**.

[0046] The shelf **24**, the one-piece ring **26**, and the susceptor **28** are arranged within the interior **42** of the reaction chamber **12**. The shelf **24** and the one-piece ring **26** are arranged within the interior **42** of the reaction chamber **12** at locations longitudinally between the injection flange **14** and the exhaust flange **16**. In this respect the shelf **24** is fixed within the interior **42** of the reaction chamber **12** at a location longitudinally adjacent to the injection flange **14**, the one-piece ring **26** abuts the shelf **24** at a lateral joint **58** located longitudinally between the shelf **24** and the one-piece ring **26**, and the shelf **24** and the one-piece ring **26** cooperatively divide the interior **42** of the reaction chamber **12** into an upper chamber **54** and a lower chamber **56**. The one-piece ring **26** further defines therethrough a one-piece ring aperture **60**, which fluidly couples the lower chamber **56** of the reaction chamber **12** with the upper chamber **54** of the reaction chamber **12**. In certain examples the shelf **24** may be formed from a transmissive material, such as quartz by way of non-limiting material. In accordance with certain examples, the one-piece ring **26** may be formed from an opaque material, such as graphite by way of non-limiting example. It is also contemplated that the one-piece ring **26** may be coated, such as with a silicon carbide coating by way of non-limiting example.

[0047] The susceptor **28** is supported within the interior **42** of the reaction chamber **12** and is configured support thereon a substrate during deposition of a film onto the substrate, e.g., during deposition of a film **62** onto a wafer **64**. In this respect it is contemplated that the susceptor **28** be supported for rotation R about a rotation axis **66** within the interior **42** of the reaction chamber **12**. The rotation axis **66** extends through the one-piece ring aperture **60**, the susceptor **28** is at least partially disposed within the one-piece ring aperture **60** such that the one-piece ring **26** extends about the susceptor **28**, and a circumferential gap **68** extending about the susceptor **28** separates the susceptor **28** from the one-piece ring **26**. In certain examples the susceptor **28** may formed from an opaque material, such as graphite by way of non-limiting example. In accordance with certain examples, the susceptor **28** may be coated, such as with a silicon carbide coating by way of the non-limiting example. It is contemplated that the susceptor **28** be configured to support the wafer **64** during deposition of the film **62** onto the wafer **64**.

[0048] The spider **30** is arranged along the rotation axis **66** and is fixed in rotation relative to the susceptor **28**. The shaft **32** is fixed in rotation relative to the spider **30** and is operatively associated with the drive module **34**. The drive module **34** is configured to rotate the shaft **32**, and therethrough the spider **30** and the susceptor **28**, about the rotation axis **66** during deposition of the film **62** onto the wafer **64**. The shaft **32**, the spider **30**, and the susceptor **28** may be as shown and described in U.S. Pat. No. 6,086,680 to Foster et al., issued on Jun. 11, 2000, the contents of which are incorporated herein by reference in its entirety.

[0049] The lift pins **36** (shown in FIG. **3**) extend through the susceptor **28** and are movable between a retracted position **70** (shown in FIG. **10**) and an extended position **72** (shown in FIG. **11**) along the rotation axis **66**. When in the retracted position **70**, the lift pins **36** may be recessed within the susceptor **28** and the wafer **64** supported on the susceptor **28**, e.g., for deposition of the film **62** onto the wafer **64**. When in the extended position **72**, the lift pins **36** may protrude from the susceptor **28** such that the lift pins **36** support the wafer **64** above the susceptor **28**, e.g., for loading of the wafer **64** onto the susceptor **28** and/or for unloading of the wafer **64** from the susceptor **28**, e.g., subsequent to deposition of the film **62** onto the wafer **64**. The lift pins **36** may be as shown and described in U.S. Patent Application Publication No. 2019/0051555 A1 to Hill et al., filed on Aug. 8, 2017, the contents of which are incorporated herein by reference in its entirety.

[0050] As has been explained above, positioning of components within the reaction chamber **12** may influence properties of the film **62** deposited onto the wafer **64**, e.g., by altering gas flow and/or thermal conditions within the reaction chamber **12** during deposition of the film **62** onto the wafer **64**. For example, position of the one-piece ring **26** relative the shelf **24** may influence gas flow and/or thermal conditions within the reaction chamber **12**. Positioning of the susceptor **28** relative to the one-piece ring **26**, both statically and during rotation R, may also influence gas flow and/or thermal conditions within the reactor. And positioning of the plurality of lift pins **36** relative to the susceptor **28** may further influence gas flow and/or thermal conditions within the reaction chamber **12**. To limit variation within the film **62** due positioning of these components, as well as others, the fixture **100** is provided.

[0051] With reference to FIG. **2**, the fixture **100** is shown. The fixture **100** is configured for determining position of a target **74** within the reaction chamber **12** of the semiconductor processing system **10** and includes a frame **102**, a leveling plate **104**, a bracket **106**, and a laser profiler **108**. The frame **102** is configured for fixation relative to the reaction chamber **12**. The leveling plate **104** is supported on the frame **102**. The bracket **106** is supported on the leveling plate **104** and overlays the frame **102**. The laser profiler **108** is suspended from the bracket **106**, overlays the leveling plate

104, and has a field of view **110**. The field of view **110** extends through the leveling plate **104** and the frame **102** to determine position of the target **74**, which is in the field of view **110** of the laser profiler **108**.

[0052] In certain examples, the fixture may include a level **112**. In such examples the level may be configured to indicate level of a mounting surface **114** relative to gravity. In accordance with certain examples, the level **112** may be a first level **112** and the fixture **100** may include a second level **116** (shown in FIG. **6**). In such examples the first level **112** may extend longitudinally along the leveling plate **104** and be configured to indicate longitudinal level of the mounting surface **114**, and the second level **116** may extend laterally across the leveling plate **104** and be configured to indicate lateral level of the mounting surface **114**. The second level **116** may be orthogonal relative to the first level **112**. Either (or both) the first level **112** and the second level **116** may be fixed relative to the mounting surface **114**, e.g., permanently affixed to the leveling plate **104**. Either (or both) the first level **112** and the second level **116** may be removably fixed to the mounting surface **114**. As will be appreciated by those of skill in the art in view of the present disclosure, fixation may simplify the setup of the fixture **100**. As will also be appreciated by those of skill in the art in view of the present disclosure, removable fixation may simplify calibration of either (or both) the first level **112** and the second level **116**.

[0053] In accordance with certain examples, the fixture **100** may include a leveling screw **118**. In such examples the leveling screw **118** spaces the leveling plate **104** from the frame **102** and is configured to control level of the leveling plate **104** relative to frame **102**, e.g., longitudinal level and/or lateral level of the leveling plate **104** relative to the frame **102**. Level control may be provided, for example, by threaded engagement of a shank of the leveling screw **118** within the leveling plate **104** and abutment of a tip of the leveling screw **118** against the frame **102**. However, as will be appreciated by those of skill in the art in view of the present disclosure, other arrangements of the leveling screw **118** are possible within the scope of the present disclosure.

[0054] It is contemplated that the leveling screw **118** may be a first leveling screw **118** and that the fixture **100** may include a second leveling screw **120**. In such examples the second leveling screw **120** may be similar to the first leveling screw **118** and additionally be lateral and/or longitudinally offset from the first leveling screw **118** on the leveling plate **104**. In such examples the second leveling screw **120** may cooperate with the first leveling screw **118** to provide control of longitudinal level in isolation of lateral level or lateral level in isolation of longitudinal level during leveling of the leveling plate **104**. Although two (2) leveling screws are shown in FIG. **2** it is to be understood and appreciated that the fixture **100** may include more than two (2) leveling screws, e.g., three (3) or more leveling screws, and remain within the scope of the present disclosure.

[0055] In certain examples, the target **74** may include the shelf **24** (shown in FIG. **1**). In such examples the laser profiler **108** may be supported above the shelf **24** to acquire height(s) of the laser profiler **108** relative to the shelf **24**. In accordance with certain examples, the target **74** may include the one-piece ring **26** (shown in FIG. **1**). In such examples the laser profiler **108** may be supported above the one-piece ring **26** to acquire height(s) of the laser profiler **108** relative to the one-piece ring **26**. In further examples, the target may include the susceptor **28**. In such examples the laser profiler **108** may be supported above the susceptor **28** such that the laser profiler **108** may acquire height(s) of the laser profiler **108** relative to the susceptor **28** and/or the one-piece ring **26** at location(s) adjacent to the susceptor **28**. It is also contemplated that the laser profiler **108** may be supported above a wafer, e.g., the wafer **64**, positioned between the susceptor **28** and the leveling plate **104** to acquire height(s) of the laser profiler **108** relative to the wafer **64**. In this respect it is contemplated that the laser profiler **108** may be supported at a location overlaying the rotation axis **66** and/or a location radially offset from the rotation axis **66**.

[0056] In certain examples, the fixture **100** may include a controller **122**. The controller **122** may be disposed in communication with the laser profiler **108**, such as by a wired link **132**. As shown in FIG. **2**, the controller **122** includes a processor **124**, a memory **126**, a device interface **128**, and a user interface **130**. The device interface **128** connects the device interface **128** with the controller **122**, e.g., via the wired link **132**. The processor **124** is operatively connected to the user interface **130** and disposed in communication with the memory **126**. The memory **126** includes a non-transitory machine-readable medium having a plurality of program modules **134** recorded thereon that, when read by the processor **124**, cause the processor **124** to execute certain operations. Among those operations are operations of a method **600** (shown in FIG. **15**) of determining position of a target, e.g., the target **74**, a reaction chamber, e.g. the reaction chamber **12**, using height(s) of the laser profiler **108** relative to the target.

[0057] It is contemplated that, in certain examples, the controller **122** be configured to determine height and/or level of the shelf **24** (shown in FIG. **1**) within the reaction chamber **12** using height(s) of the laser profiler **108** relative to the shelf **24** acquired by the laser profiler **108**. The controller **122** may be configured to determine height and/or level of the one-piece ring **26** (shown in FIG. **1**) relative to the shelf **24** using heights of the laser profiler **108** relative to the one-piece ring **26** and/or the shelf **24** acquired by the laser profiler **108**. Alternatively (or additionally), the controller **122** may be configured to determine height, level, centering, wobble, and/or runout of the susceptor **28** (shown in FIG. **1**) using height(s) of the laser profiler **108** relative to the susceptor **28** and/or the one-piece ring **26** using the laser profiler **108**. It is also contemplated that the controller **122** may be configured to determined contact height and/or level of the plurality of lift pins **36** (shown in FIG. **3**) using height(s) of the laser profiler **108** relative to a wafer, e.g., the wafer **64** (shown in FIG. **1**), supported by the plurality of lift pins **36** acquired by the laser profiler **108**. In this respect speed during movement of the plurality of lift pins **36** relative to the susceptor **28** may be tuned based on the determined contact height, e.g., by established a height at which the plurality of lift pins **36** decelerate prior to contacting a wafer supported on the susceptor **28**.

[0058] With reference to FIG. **3**, the frame **102** is shown. The frame **102** is configured to support the laser profiler **108** at a location above the target **74** (shown in FIG. **2**), e.g., the shelf **24**, the one-piece ring **26**, the susceptor **28**, and/or the lift pins **36**, and in this respect includes a first stringer **136**, a second stringer **138**, and a plurality of joists **140**. The first stringer **136** longitudinally spans the reaction chamber **12** and is supported at longitudinally opposite ends by the injection flange **14** and the exhaust flange **16**. The second stringer **138** extends in parallel with the first stringer **136**, also spans the reaction chamber **12**, and is further supported at longitudinally opposite ends by the injection flange **14** and the exhaust flange **16**. The plurality of joists **140** laterally span the reaction chamber **12** and connect the first stringer **136** to the second stringer **138**. It is contemplated that the plurality of joists **140** and at least portions of the first stringer **136** and the second stringer **138** and bound a frame aperture **142**. It is contemplated that the frame aperture **142** overlay the target **74** such that the field of view **110** of the laser profiler **108** passes through the frame aperture **142**.

[0059] With reference to FIG. **4**, the bracket **106** is shown. The bracket **106** is configured to suspend the laser profiler **108** above the leveling plate **104** while the bracket **106** is supported by the leveling plate **104**. In this this respect the bracket **106** includes a U-shaped body **144** and an end cap **146** with a cable passthrough **148**. The U-shaped body **144** has an upper tine **150**, a lower tine **152**, and a web **154**. The web **154** extends between the upper tine **150** and the lower tine **152**, and connects the upper tine **150** to the lower tine **152**. The end cap **146** is seated to ends of the lower tine **152** and the upper tine **150** located on side of the lower tine **152** and the upper tine **150** opposite the web **154**. It is contemplated that the end cap **146** be spaced apart from the web **154** by the lower tine **152** and the upper tine **150**.

[0060] The upper tine **150** of the U-shaped body **144** has a profiler seat **156** and is spaced apart from the lower tine **152** by the end cap **146** and the web **154**. The profiler seat **156** opposes the lower tine **152**, is substantially planar, and seats thereon the laser profiler **108**. More specifically, the laser profiler **108** is suspended from the bracket **106** at the profiler seat **156** of the upper tine **150**, the field of view **110** of the laser profiler **108** thereby extending toward the lower tine **152** from the profiler seat **156** of the upper tine **150** and passing through the lower tine **152.** In certain examples the laser profiler **108** may be fixed to the profiler seat **156** by one or more fasteners. In certain examples the one or more fastener may be received in the upper tine **150** and threadedly seated in the laser profiler **108**.

[0061] The lower tine **152** has a bracket seat **158** and a bracket aperture **160**. The bracket aperture **160** extends through the lower tine **152** at a location between the web **154** and the end cap **146** and is in registration with the laser profiler **108** such that the field of view **110** of the laser profiler **108** extends through the bracket aperture **160**. The bracket seat **158** is defined on a side of the lower tine **152** opposite the upper tine **150** and is substantially parallel to the profiler seat **156**. It is contemplated that the profiler seat **156** be conformal with one or mount, e.g., a shelf mount **162** (shown in FIG. **6**) defined on the mounting surface **114** of the leveling plate **104**, level of the mounting surface **114** thereby transferring the profiler seat **156**. As will be appreciated by those of skill in the art in view of the present disclosure, transferring level of the mounting surface **114** to the profiler seat **156** via the bracket seat **158** may simplify setup of the fixture **100** (shown in FIG. **1**) on the reaction chamber **12** (shown in FIG. **1**).

[0062] A cable **149** extends through the cable passthrough **148** to communicatively couple the laser profiler **108** to controller **122** (shown in FIG. **2**), e.g., via the wired or wireless link **132**. In this respect the cable passthrough **148** extends through the end cap **146** and is routed therethrough to the laser profiler **108**. In certain examples, the cable **149** may provide electrical power to the laser profiler **108**. In accordance with certain examples, the cable **149** may provide data communication to the laser profiler **108**. It is further contemplated that the cable passthrough **148** may extend through the U-shaped body **144** and remain within the scope of the present disclosure. Although shown and described herein as employed the cable **149** for power and/or data communication, it is also contemplated that the laser profiler **108** may be wirelessly connected to the controller **122** and/or employ batteries for power and remain within the scope of the present disclosure.

[0063] With reference to FIG. **5**, the laser profiler **108** is shown. The laser profiler **108** is configured to acquire height of the laser profiler **108** relative to the target **74** located within the reaction chamber **12** (shown in FIG. **1**) and in this respect includes a laser source **164**, an expansion lens **166**, collection optics **168**, and a sensor **170**. The laser source **164** is configured to illuminate the expansion lens **166** with a beam. In certain examples the laser source **164** may include a visible wavelength laser source. In accordance with certain examples, the laser source **164** may include a blue wavelength laser source. It is also contemplated that, in accordance with certain examples, that the laser source **164** may include a 405-nanometer laser source. As will be appreciated by those of skill in the art, employment of such laser sources may be relatively accurate when acquiring height using illumination projected through transparent structures having complex geometries, such as reaction chambers having external ribs.

[0064] The expansion lens **166** is configured to expand and project the beam received from the laser source **164** and incident upon the expansion lens **166**. More specifically, the expansion lens **166** is configured widen the beam along a singular diameter of the beam to define the line **76** using the beam incident upon the expansion lens **166**. It is contemplated that the laser profiler **108** project the line **76** onto the target **74** according to the location of the mount supporting the bracket **106**, e.g., the shelf mount **162**, the target in turn at least partially reflecting the line **76** and thereby coupling the expansion lens **166** to the collection optics **168**.

[0065] The collection optics **168** are configured to collect illumination reflected from the target **74** and communicate the collected illumination to the sensor **170**. The sensor **170** is configured to convert illumination received from the collection optics **168** into height(s) of the laser profiler **108** relative to the line **76**. In certain examples, the sensor is configured to convert the illumination received from the collection optics **168** into a first height A of the laser profiler **108** above a first point **50** along the line **76** on the target **74** and a second height B of the laser profiler **108** above a second point **52** along the line **76** on the target **74**, the second height B being spaced apart from the first height A along the line **76**. In such examples, the sensor **170** may further communicate the first height A and the second height B to the controller **122**, e.g., for determining position of the target **74** within the reaction chamber **12** (shown in FIG. **1**) and/or to display the heights on user interface **130**.

[0066] With reference to FIG. **6**, the leveling plate **104** is shown. In the illustrated example the leveling plate **104** includes the shelf mount **162**. The shelf mount **162** is located on the mounting surface **114** of the leveling plate **104** and includes a shelf mount aperture **172** and a shelf mount registration feature **174**. The shelf mount aperture **172** extends through the leveling plate **104** at a location that, when the leveling plate **104** is supported on the frame **102** and the frame **102** supported on the injection flange **14** and the exhaust flange **16**, overlays the shelf **24**. The shelf mount registration feature **174** is located on the mounting surface **114** at a location adjacent to the shelf mount aperture **172** and is configured to fix the bracket **106** on the mounting surface **114** such that at least a portion of the shelf **24** is in the field of view **110** (shown in FIG. **4**) of the laser profiler **108**.

[0067] It is contemplated that the laser profiler **108** (shown in FIG. **2**) be seated on the shelf mount **162** and project the line **76** (shown in FIG. **5**) onto the shelf **24** through the leveling plate **104** and the frame **102** at the shelf mount **162**. It is further contemplated that the laser profiler **108** acquire height of the laser profiler **108** relative to the shelf **24** at the shelf mount **162** along the line **76**. The acquired height may be communicated to the controller **122** (shown in FIG. **2**), and the controller **122** may determine position of the shelf **24** using the height received from the laser profiler **108**. In certain examples, determining position of the shelf **24** may include determining height and/or longitudinal level of the shelf **24** relative to the mounting surface **114** of the leveling plate **104**.

[0068] In certain examples, the shelf mount **162** may be a first shelf mount **162** and the leveling plate **104** may include a second shelf mount **176**. In such examples, the second shelf mount **176** may be located on the mounting surface **114** at a location laterally offset from the first shelf mount **162**. In such examples, the second shelf mount **176** may include a second shelf mount aperture **178** and a second shelf mount registration feature **180**. The second shelf mount aperture **178** may extend through the leveling plate **104** at a location overlaying the shelf **24** laterally offset from the first shelf mount **162**. The second shelf mount registration feature **180** may be located on the mounting surface **114** at a location adjacent to the second shelf mount aperture **178**. The second shelf mount registration feature **180** may further be configured to fix the bracket **106** on the mounting surface **114** such that a second portion of the shelf **24** is in the field of view **110** of the laser profiler **108**.

[0069] In accordance with certain examples, the bracket **106** (shown in FIG. **2**) and the laser profiler **108** (shown in FIG. **2**) may be removed from the first shelf mount **162** and seated on the second shelf mount **176** once position of the shelf **24** has been determined by the controller **122** (shown in FIG. **2**). Once seated on the second shelf mount **176**, the laser profiler **108** may project the line **76** onto a second portion of the shelf **24** and acquire height of the laser profiler **108** relative to the second portion of the shelf **24** along the line **76**. The acquired height may be communicated to the controller **122**, and the controller **122** may determine additional positional information of the shelf **24**. In certain examples, the additional positional information may include lateral level of the shelf **24** relative to the mounting surface **114** of the leveling plate **104**.

[0070] With reference to FIG. **7**, the fixture **100** is shown according to an example including a leveling plate **204**. The leveling plate **204** has a mounting surface **214**, is similar to the leveling plate **104** (shown in FIG. **2**), and additionally includes a one-piece ring mount **262**. The one-piece ring mount **262** is located on the mounting surface **214** and includes a one-piece ring mount aperture **272** and a one-piece ring mount registration feature **274**. The one-piece ring mount aperture **272** extends through the leveling plate **204** at a location that, when the leveling plate **104** is supported on the frame **102**, and the frame **102** supported on the injection flange **14** and the exhaust flange **16**, the one-piece ring mount aperture **272** overlays the one-piece ring **26**. The one-piece ring mount registration feature **274** is located on the mounting surface **214** at a location adjacent to the one-piece ring mount aperture **272** and is configured to fix the bracket **106** (shown in FIG. **2**) on the mounting surface **214** such that a portion of the one-piece ring **26** is in the field of view **110** (shown in FIG. **2**) of the laser profiler **108** (shown in FIG. **2**). In certain examples, the one-piece ring mount **262** may be longitudinally and/or laterally offset from the shelf mount **162** (shown in FIG. **6**).

[0071] It is contemplated that the laser profiler **108** (shown in FIG. **2**) be seated on the one-piece ring mount **262** and project the line **76** (shown in FIG. **5**) onto the one-piece ring **26** through the leveling plate **204** and the frame **102** at the one-piece ring mount **262**. It is further contemplated that the laser profiler **108** acquire height of the laser profiler **108** relative to the one-piece ring **26** at the one-piece ring mount **262**. The acquired height may be communicated to the controller **122** (shown in FIG. **2**), and the controller **122** may determine position of the one-piece ring **26** using the height received from the laser profiler **108**. In certain examples, determining position of the one-piece ring **26** may include determining height and/or longitudinal level of the one-piece ring **26** relative to the mounting surface **214**.

[0072] In certain examples, the one-piece ring mount **262** may be a first one-piece ring mount **262** and the leveling plate **204** may include a second one-piece ring mount **276**. In such examples, the second one-piece ring mount **276** may be located on the mounting surface **214** at a location laterally offset from the first one-piece ring mount **262**. In such examples, the second one-piece ring mount **276** may include both a second one-piece ring mount aperture **278** and a second one-piece ring mount registration feature **280**. The second one-piece ring mount aperture **278** may extend through the leveling plate **104** at a location overlaying the one-piece ring **26** and laterally offset from the first one-piece ring mount **262**. The second one-piece ring mount registration feature **280** may be located on the mounting surface **214** at a location adjacent to the second one-piece ring mount aperture **278**, and may further be configured to fix the bracket **106** on the mounting surface **214** of the leveling plate **204** such that a second portion of the one-piece ring **26** is in the field of view **110** (shown in FIG. **2**) of the laser profiler **108** (shown in FIG. **2**).

[0073] In accordance with certain examples, the bracket **106** (shown in FIG. **2**) and the laser profiler **108** (shown in FIG. **2**) may be removed from the first one-piece ring mount **262** once the heights are acquired at the first susceptor mount **362**, and thereafter be seated on the second one-piece ring mount **276**. Once seated on the second one-piece ring mount **276**, the laser profiler **108** may project the line **76** (shown in FIG. **5**) onto a second portion of the one-piece ring **26** and acquire height of the laser profiler **108** relative to the second portion of the one-piece ring **26** along the line **76**. The acquired height may be communicated to the controller **122** (shown in FIG. **2**), and the controller **122** may determine additional position information of the one-piece ring **26** using the heights acquired from the second one-piece ring mount **276**. In certain examples, the additional position information may include lateral level of the one-piece ring **26** relative to the mounting surface **214** of the leveling plate **204**.

[0074] With reference to FIG. **8**, the fixture **100** is shown according to an example including a leveling plate **304**. The leveling plate **304** has a mounting surface **314**, is similar to the leveling plate **104** (shown in FIG. **2**), and additionally includes a susceptor mount **362**. The susceptor mount **362** is located on the mounting surface **314** and includes a susceptor mount aperture **372** and a susceptor mount registration feature **374**. The susceptor mount aperture **372** extends through the leveling plate **304** at a location that, when the leveling plate **304** is supported on the frame **102**, and the frame **102** supported on the injection flange **14** and the exhaust flange **16**, the susceptor mount aperture **372** overlays the susceptor **28**. The susceptor mount registration feature **374** is located on the mounting surface **314** at a location adjacent to the susceptor mount aperture **372** and is configured to fix the bracket **106** (shown in FIG. **2**) on the mounting surface **314** such that a portion of the susceptor **28** is in the field of view **110** (shown in FIG. **2**) of the laser profiler **108** (shown in FIG. **2**). In certain examples, the susceptor mount **362** may be longitudinally and/or laterally offset from the one-piece ring mount **262** (shown in FIG. **7**).

[0075] It is contemplated that the laser profiler **108** (shown in FIG. **2**) be seated on the susceptor mount **362** and project the line **76** (shown in FIG. **5**) onto the susceptor **28** through the leveling plate **304** and the frame **102** from the susceptor mount **362**. It is further contemplated that the laser profiler **108** acquire height of the laser profiler **108** relative to the susceptor **28** at the susceptor mount **362**. The acquired height may be communicated to the controller **122** (shown in FIG. **2**), and the controller **122** may determine position of the susceptor **28** using the height received from the laser profiler **108**. In certain examples, determining position of the susceptor **28** may include determining height and/or longitudinal level of the susceptor **28** relative to the mounting surface **314** of the leveling plate **304**.

[0076] It is contemplated that the line **76** (shown in FIG. **5**) may be projected through the susceptor mount aperture **372** onto a portion of the susceptor **28** bounding the circumferential gap **68**, a portion of the one-piece ring **26** bounding the circumferential gap **68**, and into the circumferential gap **68**. In such examples the first point **50** and the second point **52** may be located at edges of the susceptor **28** and the one-piece ring **26** bounding the circumferential gap **68**, and width **78** acquired by determining width of the circumferential gap **68** between the first point **50** and the second point **52**. As will be appreciated by those of skill in the art in view of the present disclosure, the width **78** may be determined without imaging the circumferential gap, limiting error associated with imaging the circumferential gap **68** and portions of the susceptor **28** and the one-piece ring **26** bounding the circumferential gap **68**. In certain examples, height measurements acquired by the laser profiler **108** (shown in FIG. **2**) may be displayed on the user interface **130**, simplifying analysis of the width **78** by a user.

[0077] In certain examples, the susceptor mount **362** may be a first susceptor mount **362** and the leveling plate **304** may include a second susceptor mount **376**. In such examples the second susceptor mount **376** may be located on the mounting surface **314** at a location laterally offset from the first susceptor mount **362** and may include both a second susceptor mount aperture **378** and a second susceptor mount registration feature **380**. The second susceptor mount aperture **378** may extend through the leveling plate **304** at a location overlaying the susceptor **28** and laterally offset from the first susceptor mount **362**. The second susceptor mount registration feature **380** may be located on the mounting surface **314** at a location adjacent to the second susceptor mount aperture **378**. The second susceptor mount registration feature **380** may further be configured to fix the bracket **106** (shown in FIG. **2**) on the mounting surface **314** such that a second portion of the susceptor **28** is in the field of view **110** (shown in FIG. **2**) of the laser profiler **108**

[0078] Once height of the laser profiler **108** relative to the susceptor **28** is acquired from the first susceptor mount **362**, the bracket **106** and the laser profiler **108** may be removed from the first susceptor mount **362** the bracket **106** and the laser profiler **108** seated at the second susceptor mount **376**. Once seated on the second susceptor mount **376**, the laser profiler **108** may project the line **76** (shown in FIG. **5**) onto a second portion of the susceptor **28** and acquire height of the laser profiler **108** relative to the second portion of the susceptor **28** along the line **76**. The acquired height may be communicated to the controller **122** (shown in FIG. **2**), and the controller **122** may determine additional position information of the susceptor **28** using the heights acquired from the second susceptor mount **376**. In certain examples, the additional position information may include lateral level of the susceptor **28** relative to the mounting surface **314** of the leveling plate **304**.

[0079] In accordance with certain examples, determining position of the susceptor **28** may include determining wobble of the susceptor **28** using heights acquired by the laser profiler **108** (shown in FIG. **2**). In this respect the susceptor **28** may be rotated R about the rotation axis **66** while the laser profiler **108** is supported at susceptor mount **362**. As the susceptor **28** rotates it is contemplated that the laser profiler **108** acquire height of the laser profiler **108** relative to the susceptor **28** at two or more rotary positions about the rotation axis **66**, and the controller **122** determines wobble of the susceptor **28** by determining a height differential between heights of the susceptor **28** at the two or more rotary positions. In certain examples, wobble of the susceptor **28** may be displayed on the user interface **130** by change in a portion of the line **76** displaying height of the laser profiler **108** above the susceptor **28** during rotation R of the susceptor **28** about the rotation axis **66**, simplifying analysis of wobble of the susceptor **28** by a user. The wobble may be determined continuously during rotation of the susceptor **28** about the rotation axis **66**, improving wobble determination through acquisition of a differential in the width **78** of the circumferential gap **68** during 360-degrees or more of rotation about rotation axis **66**.

[0080] In further examples, determining position of the susceptor **28** may include determining centering of the susceptor **28** using heights acquired by the laser profiler **108** (shown in FIG. **2**). In this respect the differential in the width **78** of the circumferential gap **68** may be determined with the susceptor **28** in the both the first rotary position and the second rotary position. Centering may be determined using the differential in the width **78** at both the first rotary position and the second rotary position. In certain examples, gap width may be determined while the susceptor is rotated about the rotation axis **66** and runout determined from the gap width during rotation R of the susceptor about the rotation axis **66**. In further examples, runout of the susceptor **28** may be displayed on the user interface **130** by change in a portion of the line **76** projected into the circumferential gap **68** during rotation R of the susceptor **28** about the rotation axis **66**, simplifying analysis of runout of the susceptor **28** by a user. Runout of the susceptor **28** may be determined continuously during rotation of the susceptor **28** about the rotation axis **66**, improving runout determination through acquisition of the differential in the width **78** of the circumferential gap **68** by 360-degrees or more of rotation about rotation axis **66**.

[0081] With reference to FIGS. **9-11**, the fixture **100** is shown according to an example including a leveling plate **404**. As shown in FIG. **9**, the leveling plate **404** is similar to the leveling plate **104** (shown in FIG. **2**) and additionally has a mounting surface **414** with a wafer center mount **462**. The wafer center mount **462** is configured for determining contact height of the plurality of lift pins **36** and in this respect includes a wafer center mount aperture **472** and a wafer center mount registration feature **474**.

[0082] The wafer center mount registration feature **474** is configured to fix the bracket **106** on the mounting surface **414** and is located on the mounting surface **414** at a location adjacent to the wafer center mount aperture **472**. The wafer center mount aperture **472** extends through the leveling plate **404** at a location that, when the leveling plate **404** is supported on the frame **102** and overlays the reaction chamber **12**, is registered to the rotation axis **66**. The registration is such that seating the bracket **106** (shown in FIG. **2**) on the wafer center mount **462** causes the field of view **110** (shown in FIG. **2**) of the laser profiler **108** (shown in FIG. **2**) to extend into the reaction chamber **12** through both the leveling plate **404** and the frame **102**. It is further contemplated that the field of view **110** overlay a central portion of a wafer seated on the susceptor **28**, e.g., a test wafer **480**, such that the laser profiler **108** may acquire height of the laser profiler **108** relative to the test wafer **480** to determine contact height of the lift pins **36**.

[0083] As shown in FIG. **10**, to determine lift pin contact height, the plurality of lift pins **36** are first moved from the extended position **72** to the retracted position **70**. Movement of the lift pins **36** to the retracted position **70** seats the test wafer **480** on the susceptor **28** such that the center portion of the test wafer **480** is in the field of view **110** of the laser profiler **108**. Once the test wafer **480** is seated on the susceptor **28**, the laser profiler **108** projects the line **76** onto the center portion of the test wafer **480** and acquires height of the laser profiler **108** relative to the center portion of the test wafer **480**. Height of the laser profiler **108** relative to the center portion of the test wafer **480** is communicated to the controller **122** (shown in FIG. **2**), which determines position of the test wafer **480** using the height acquired by the laser profiler **108**. In certain examples, the laser profiler **108** may acquire a plurality of heights continuously, e.g., sequentially and separated by a predetermined time interval. It is contemplated that the controller **122** in turn determine contact height by comparing heights received from the laser profiler **108** with a prior-acquired height and determine contact height according to differential between an acquired height and the prior-acquired height.

[0084] As shown in FIG. **11**, the lift pins **36** are next driven from the retracted position **70** toward the extended position **72**. As will be appreciated by those of skill in the art in view of the present disclosure, driving the lift pins **36** toward the extended position **72** causes the lift pins **36** to displace the test wafer **480** from the susceptor **28** when movement of the lift pins **36** is such that the lift pins **36** come into contact the underside of the test wafer **480**. It is contemplated that the controller **122** (shown in FIG. **2**) determine the lift pin height at which the lift pins **36** contact the wafer **480** by comparing each successive height measurement received from the laser profiler **108** with the prior-acquired height comparing height measurement received after the lift pins **36** contact the underside of the test wafer **480**.
[0085] In certain examples, determined contact height may be used to control movement of the lift pins during placement of wafers thereafter on the susceptor **28**. For example, speed of the lift pins **36** may thereafter be slowed during movement from the extended position **72** to the retracted position **70** as the lift pins approach the lift pin contact height, limiting (or eliminating) the tendency wafers, e.g., the wafer **64**, to otherwise decenter during seating against the susceptor **28** during loading. Alternatively (or additionally), speed of the lift pins **36** may be slowed during movement of the lift pins **36** during movement from the retracted position **70** to the extended position **72**, limiting (or eliminating) the risk of wafer damage, e.g., the wafer **64** (shown in FIG. **1**), due to lift pin contact during unloading of wafers from the susceptor **28**. Advantageously, the determined lift pin contact height may be relatively accurate due to the use of the line **76**, e.g., by acquiring height of laser profiler **108** above the center of the test wafer **480** at opposite ends of the line **76** in the field of view **110** of the laser profiler **108**, in examples where level of the lift pins **36** differs from level of the susceptor **28**.
[0086] With reference to FIGS. **12-14**, the fixture **100** is shown according to an example including a leveling plate **504**. As shown in FIG. **12**, the leveling plate **504** is similar to the leveling plate **104** (shown in FIG. **2**) and additionally has a mounting surface **514** with a first wafer periphery mount **562** and at least one second wafer periphery mount **576**. The first wafer periphery mount **562** and the second wafer periphery mount **576** are each configured for determining level of the plurality of lift pins **36** using measurements of height of the laser profiler **108** (shown in FIG. **2**) relative to a periphery of a wafer, e.g., the test wafer **480**, positioned within the reaction chamber **12** between the susceptor **28** and the laser profiler **108**.
[0087] The first wafer periphery mount **562** includes a first wafer periphery mount aperture **572** and a first wafer periphery mount registration feature **574**. The first wafer periphery mount registration feature **574** is configured to register and fix the bracket **106** (shown in FIG. **2**) on the mounting surface **514** and is adjacent to the first wafer periphery mount aperture **572**. The first wafer periphery mount aperture **572** extends through the leveling plate **504** at a location that, when the leveling plate **504** is supported on the frame **102** above the reaction chamber **12**, overlays a periphery of the test wafer **480**. Registration of the first wafer periphery mount aperture **572** with the periphery of the wafer **480** causes the periphery the test wafer **480** to be in the field of view **110** (shown in FIG. **2**) of the laser profiler **108** (shown in FIG. **2**) when the bracket **106** is seated on the first wafer periphery mount **562**. Placement of the periphery of the test wafer **480** in the field of view **110** of the laser profiler **108** enables the laser profiler **108** to acquire height measurements of the periphery of the test wafer **480** from the first wafer periphery mount **562** by projecting the line **76** (shown in FIG. **5**) onto the periphery of the test wafer **480**.
[0088] The second wafer periphery mount **576** is similar to the first wafer periphery mount **562** and is additionally offset from the first wafer periphery mount **562** about the rotation axis **66**. In this respect the second wafer periphery mount registration feature **580** is configured to register and fix the bracket **106** (shown in FIG. **2**) on the mounting surface **514** at a location adjacent to the second wafer periphery mount aperture **578**. The second wafer periphery mount aperture **578** extends through the leveling plate **504** at a location that, when the leveling plate **504** is supported on the frame **102** above the reaction chamber **12**, overlays the periphery of the test wafer **480** at a location

circumferentially offset from the first wafer periphery mount aperture **572** about the rotation axis **66**. Registration of the second wafer periphery mount aperture **578** with the periphery of the test wafer **480** also causes the periphery of the test wafer **480** to be in the field of view **110** (shown in FIG. **2**) of the laser profiler **108** (shown in FIG. **2**) when the bracket **106** is seated on the second wafer periphery mount **576**. As will be appreciated by those of skill in the art in view of the present disclosure, placement of the periphery of the test wafer **480** in the field of view **110** of the laser profiler **108** also enables the laser profiler **108** to acquire height measurements of the periphery of the test wafer **480** from the second wafer periphery mount **576** by projecting the line **76** onto the periphery of the test wafer **480**.

[0089] As shown in FIG. **13**, to determine level of the plurality of lift pins **36**, the test wafer **480** is supported on the susceptor **28** and the bracket **106** seated on the first wafer periphery mount **562**. Next, the lift pins **36** driven from the retracted position **70** to the extended position **72**. As the lift pins **36** move between the retracted position **70** and the extended position **72** the laser profiler **108** continuously acquires height measurements of the laser profiler **108** relative to the periphery of the test wafer **480** from the first wafer periphery mount **562**. As the lift pins **36** contact the underside of the test wafer **480** and displace the test wafer **480** from the susceptor **28** the height measurements acquired by the laser profiler **108** change, height measurements thereafter becoming smaller as the lift pins reach the extended position **72**. In certain examples, the height measurements are communicated to the controller **122**, and the lift pin contact height reflected in the change in height identified as a first lift pin contact height.

[0090] As shown in FIG. **14**, the lift pins **36** are next returned to the retracted position **70** such that the test wafer **480** is again seated on the susceptor **28**, the bracket **106** (shown in FIG. **2**) removed from the first wafer periphery mount **562** and seated on the second wafer periphery mount **576**, and the lift pins **36** again driven from the retracted position **70** and the extended position **72**. As the lift pins **36** are driven between the retracted position **70** and the extended position **72** the laser profiler **108** acquires height measurements of the laser profiler **108** relative to the periphery of the test wafer **480** from the second wafer periphery mount **576**. As above, it is contemplated that the height measurements be communicated to the controller **122**, and that the controller **122** identify the lift pin contact height at which the acquired height measurements change as a second lift pin contact height. As will be appreciated by those of skill in the art in view of the present disclosure, the second lift pin contact height may differ from the first lift pin contact height according difference in level between the mounting surface **514** and the plurality of lift pins **36**.

[0091] To determine lift pin level, the first lift pin contact height is compared to the second lift pin contact height, e.g. by calculating a difference between the first lift pin contact height and the second lift pin contact height, and position of the lift pins **36** be determined with respect to level. In certain examples, the lift pin level may be compared to a predetermined lift pin level limit. In accordance with certain examples, level of the lift pins **36** may be adjusted when the lift pin level exceeds the predetermined lift pin level. As will be appreciated by those of skill in the art in view of the present disclosure, limiting deviation of the level of the lift pins **36** from level of the susceptor **28** may limit (or eliminate) the tendency of the level of the lift pins **36** to decenter wafers during seating on the susceptor **28**, e.g., the wafer **64** (shown in FIG. **1**), reducing (or eliminating) variation otherwise associated with such decentering.

[0092] In certain examples, the second wafer periphery mount **576** may be circumferentially offset from the first wafer periphery mount **562** about the rotation axis **66** by 180-degrees. In accordance with certain examples, the second wafer periphery mount **576** may be circumferentially offset from the first wafer periphery mount **562** about the rotation axis **66** by 90-degrees. Although shown and described herein as having the first wafer periphery mount **562** and the second wafer periphery mount **576**, it is to be understood and appreciated that examples of the mounting plate may have more than two (2) wafer periphery mounts, e.g., four (4) wafer periphery mounts distributed circumferentially about the rotation axis and remain within the scope of the present disclosure. In such examples lift pin contact heights may be acquired at each of the wafer periphery mounts, a lift pin contact height total indicated range (TIR) determined using the lift pin contact heights, and the lift pin contact height TIR compared with the predetermined lift pin limit for purposes of determining level of the plurality of lift pins **36**.

[0093] With reference to FIG. **15**, the method **600** of determining position of a target within a reaction chamber, e.g., position of the target **74** (shown in FIG. **2**) of the reaction chamber **12** (shown in FIG. **1**), is shown. The method **600** includes leveling a leveling plate supported above the reaction chamber relative to gravity, e.g., the leveling plate **104** (shown in FIG. **2**), as shown with box **610**. The method **600** also includes projecting a line, e.g., the line **76** (shown in FIG. **5**), into the reaction into the reaction chamber and onto the target using a laser profiler, e.g., the laser profiler **108** (shown in FIG. **2**), as shown with box **620**. Height of the laser profiler about the target is acquired at a first point and a second point along the line, e.g., the first point **50** (shown in FIG. **5**) and the second point **52** (shown in FIG. **5**), as shown with box **632**. It is contemplated that the second point be spaced apart from the first point along the line.

[0094] As shown with box **640**, position of the target is determined using the height of the profiler relative to the target. In certain examples, height and/or level of a shelf located within the reaction chamber may be determined, e.g., height and/or level of the shelf **24** (shown in FIG. **1**), as shown with box **642**. In accordance with certain examples, height and/or level of a one-piece ring may be determined, e.g. height and/or level of the one-piece ring **26** (shown in FIG. **1**), as shown with box **644**. In further examples, one or more of height, level, centering, wobble, and/or runout of a susceptor may be determined using the heights acquired by the laser profiler, e.g., height, level, and centering of the susceptor **28** within the one-piece ring **26**, as shown with box **646**. Wobble and/or runout of the susceptor **28** (shown in FIG. **1**) may further be determined using the heights acquired by the laser profiler, as further shown with box **646**. It is also contemplated that either of lift pin contact height and/or level may be determined using the heights acquired by the laser profiler, e.g., contact height and/or level of the lift pins **36** (shown in FIG. **1**), as shown with box **648**.

[0095] As shown with box **650**, the determined position of the target may be compared to a predetermined target limit. It is contemplated that the target position may be accepted when the position is within the predetermined limit, as shown with arrow **680** and box **670**. It is also contemplated that the target may be adjusted when the position of the target exceeds the predetermined limit, as shown with box **660**. The position of the target and/or of other targets may be checked subsequent to adjustment of the target, as shown with arrow **690**.

[0096] The particular implementations shown and described are illustrative of the invention and its best mode and are not intended to otherwise limit the scope of the aspects and implementations in any way. Indeed, for the sake of brevity, conventional manufacturing, connection, preparation, and other functional aspects of the system may not be described in detail. Furthermore, the connecting lines shown in the various figures are intended to represent exemplary functional relationships and/or physical couplings between the various elements. Many alternative or additional functional relationship or physical connections may be present in the practical system, and/or may be absent in some embodiments.

[0097] It is to be understood that the configurations and/or approaches described herein are exemplary in nature, and that these specific embodiments or examples are not to be considered in a limiting sense, because numerous variations are possible. The specific routines or methods described herein may represent one or more of any number of processing strategies. Thus, the various acts illustrated may be performed in the sequence illustrated, in other sequences, or omitted in some cases.

[0098] The subject matter of the present disclosure includes all novel and nonobvious combinations and subcombinations of the various processes, systems, and configurations, and other features, functions, acts, and/or properties disclosed herein, as well as any and all equivalents thereof.

ELEMENT LISTING

[0099] **10** Semiconductor Processing System [0100] **12** Reactor [0101] **14** Injection Flange [0102] **16** Exhaust Flange [0103] **18** First Precursor Source [0104] **20** Second Precursor Source [0105] **22** Purge Source [0106] **24** Shelf [0107] **26** One-Piece Ring [0108] **28** Susceptor [0109] **30** Spider [0110] **32** Shaft [0111] **34** Drive Module [0112] **36** Lift Pins [0113] **38** Injection end [0114] **40** Exhaust End [0115] **42** Interior [0116] **44** First Precursor [0117] **46** Second Precursor [0118] **48** Purge Gas [0119] **50** First Point [0120] **52** Second Point [0121] **54** Upper Chamber [0122] **56** Lower Chamber [0123] **58** Lateral Joint [0124] **60** One-Piece Ring Aperture [0125] **62** Film [0126] **64** Wafer [0127] **66** Rotation Axis [0128] **68** Circumferential Gap [0129] **70** Retracted Position [0130] **72** Extended Position [0131] **74** Target [0132] **76** Line [0133] **78** Width of Circumferential Gap [0134] **80** First Rotary Position [0135] **82** Second Rotary Position [0136] **100** Fixture [0137] **102** Frame [0138] **104** Leveling Plate [0139] **106** Bracket [0140] **108** Laser Profiler [0141] **110** Field of View [0142] **112** First Level [0143] **114** Mounting Surface [0144] **116** Second Level [0145] **118** First Leveling Screw [0146] **120** Second Leveling Screw [0147] **122** Controller [0148] **124** Processor [0149] **126** Memory [0150] **128** Device Interface [0151] **130** User Interface [0152] **132** Wired Link [0153] **134** Program Modules [0154] **136** First Stringer [0155] **138** Second Stringer [0156] **140** Plurality of Joists [0157] **142** Frame Aperture [0158] **144** U-Shaped Body [0159] **146** End Cap [0160] **148** Cable Passthrough [0161] **149** Cable [0162] **150** Upper Tine [0163] **152** Lower Tine [0164] **154** Web [0165] **156** Profiler Seat [0166] **158** Bracket Seat [0167] **160** Bracket Aperture [0168] **162** First Shelf Mount [0169] **164** Laser Source [0170] **166** Expansion Lens [0171] **168** Collection Optics [0172] **170** Sensor [0173] **172** First Shelf Mount Aperture [0174] **174** First Shelf Mount Registration Feature [0175] **176** Second Shelf Mount [0176] **178** Second Shelf Mount Aperture [0177] **180** Second Shelf Mount Registration Feature [0178] **204** Leveling Plate [0179] **214** Mounting Surface [0180] **262** First One-Piece Ring Mount [0181] **272** First One-Piece Ring Mount Aperture [0182] **274** First One-Piece Ring Mount Registration Feature [0183] **276** Second One-Piece Ring Mount [0184] **278** Second One-Piece Ring Mount Aperture [0185] **280** Second One-Piece Ring Mount Registration Feature [0186] **304** Leveling Plate [0187] **314** Mounting Surface [0188] **362** First Susceptor Mount [0189] **372** First Susceptor Mount Aperture [0190] **374** First Susceptor Mount Registration Feature [0191] **376** Second Susceptor Mount [0192] **378** Second Susceptor Mount Aperture [0193] **380** Second Susceptor Mount Registration Feature [0194] **404** Leveling Plate [0195] **414** Mounting Surface [0196] **462** Wafer Center Mount [0197] **472** Wafer Center Mount Aperture [0198] **474** Wafer Center Mount Registration Feature [0199] **480** Test Wafer [0200] **504** Leveling Plate [0201] **514** Mounting Surface [0202] **562** First Wafer Periphery Mount [0203] **572** First Wafer Periphery Mount Aperture [0204] **574** First Wafer Periphery Mount Registration Feature [0205] **576** Second Wafer Periphery Mount [0206] **578** Second Wafer Periphery Mount Aperture [0207] **580** Second Wafer Periphery Mount Registration Feature [0208] **600** Method [0209] **610** Box [0210] **620** Box [0211] **622** Box [0212] **624** Box [0213] **626** Box [0214] **628** Box [0215] **630** Box [0216] **632** Box [0217] **640** Box [0218] **642** Box [0219] **644** Box [0220] **646** Box [0221] **648** Box [0222] **650** Box [0223] **660** Box [0224] **670** Box [0225] **680** Arrow [0226] **690** Arrow

Claims

1. A method of determining a position of a target within a reaction chamber of a semiconductor processing system, comprising: at a fixture including a frame fixed above a reaction chamber, a leveling plate supported on the frame, a bracket supported on the leveling plate, and a laser profiler suspended from the bracket and overlaying the reaction chamber, the laser profiler having a field of view extending through the leveling plate and the frame, leveling the leveling plate above the reaction chamber; projecting a line into the reaction chamber and onto a target within the reaction chamber; acquiring a height of the laser profiler above the target at a first point and a second point along the line, the second point spaced apart from the first point along the line; and determining the

position of the target using the height of the laser profiler above the target at the first point and the second point along the line.
2. The method of claim 1, wherein the target comprises a shelf fixed within an interior of the reaction chamber, wherein determining the position of the target comprises determining at least one of a height and a level of the shelf relative to the leveling plate.
3. The method of claim 1, wherein the target comprises a one-piece ring seated within the reaction chamber, wherein determining the position of the target comprises determining at least one of a height and a level of the one-piece ring relative to the leveling plate.
4. The method of claim 1, wherein the target comprises a susceptor supported for rotation about a rotation axis within the reaction chamber, wherein determining the position of the target comprises determining at least one of (a) a height of the susceptor relative to the leveling plate, (b) a level of the susceptor relative to the leveling plate, (c) a centering of the susceptor relative to a one-piece ring extending about the susceptor, (d) a wobble of the susceptor during rotation about the rotation axis, and (e) a runout of the susceptor during rotation about the rotation axis.
5. The method of claim 1, further comprising: supporting a wafer on a susceptor supported within the reaction chamber for rotation about a rotation axis; and displacing the wafer from the susceptor using a plurality of lift pins translatable along the rotation axis relative to the susceptor, wherein the projecting the line onto the target comprises projecting the line onto the wafer, and wherein determining the position of the target comprises identifying a contact height of the plurality lift pins with the wafer prior to displacement of the wafer from the susceptor.
6. The method of claim 5, wherein projecting the line onto the target comprises projecting the line onto a center portion of the wafer, the method further comprising tuning a speed of movement of the plurality of lift pins between an extended position and a retracted position along the rotation axis during loading of wafers on the susceptor.
7. The method of claim 5, wherein the projecting the line onto the target comprises projecting the line onto a peripheral portion of the wafer, the method further comprising determining a level of the plurality of lift pins relative to the leveling plate using the contact height of the plurality of lift pins.
8. The method of claim 7, wherein the contact height is a first contact height, the method further comprising acquiring a second contact height of the plurality of lift pins, wherein level of the plurality of lift pins is determined using both the first contact height and the second contact height of the plurality of lift pins.
9. A method of determining a position of a target within a reaction chamber of a semiconductor processing system, comprising: at a fixture including a frame fixed above a reaction chamber, a leveling plate supported on the frame, a bracket supported on the leveling plate, and a laser profiler coupled to the bracket and overlaying the reaction chamber, the laser profiler having a field of view extending through the leveling plate and the frame, and the frame including a first string, a second stringer offset from the first stringer, and a plurality of joists extending between the first stringer and the second stringer, leveling the leveling plate above the reaction chamber; projecting a line from the laser profiler into the reaction chamber and onto a target within the reaction chamber; acquiring a height of the laser profiler above the target at a first point and a second point along the line, the second point spaced apart from the first point along the line; and determining the position of the target using the height of the laser profiler above the target at the first point and the second point along the line.
10. The method of claim 9, further comprising comparing the determined position of the target to a predetermined target limit.
11. The method of claim 10, further comprising accepting the determined position of the target in response to the determined position being within the predetermined target limit.
12. The method of claim 10, further comprising adjusting the target in response to the determined position of the target being outside the predetermined target limit.
13. The method of claim 12, wherein the leveling the level plate comprises adjusting a first leveling

screw, wherein the first leveling screw is threaded through the leveling plate and a tip of the first leveling screw is disposed against the frame, such that the first leveling screw spaces the leveling plate from the frame.

14. The method of claim 13, wherein the leveling the level plate further comprises adjusting a second leveling screw, wherein the second leveling screw is threaded through the leveling plate and a tip of the second leveling screw is disposed against the frame, such that the second leveling screw spaces the leveling plate from the frame.

15. The method of claim 14, wherein the first leveling screw and the second leveling screw operate to provide control of at least one of a longitudinal level or a lateral level of the leveling plate.

16. A method of determining a position of a target within a reaction chamber of a semiconductor processing system, comprising: leveling a leveling plate above the reaction chamber, wherein a first level coupled to the leveling plate and extending longitudinally along a mounting surface of the leveling plate and a second level coupled to the leveling plate and extending laterally along the mounting surface of the leveling plate are used for the leveling the leveling plate; projecting a line into the reaction chamber and onto a target from a laser profiler coupled to a bracket supported on the leveling plate; acquiring a height of the laser profiler above the target at a first point and a second point along the line; and determining the position of the target using the height of the laser profiler above the target at the first point and the second point along the line.

17. The method of claim 16, wherein the projecting the line from the laser profiler comprises: emitting a beam from a laser source comprised in the laser profiler to an expansion lens comprised in the laser profiler; and expanding the beam into the line on the target.

18. The method of claim 17, further comprising: collecting, via a collection optic comprised in the laser profiler, illumination from the line reflected from the target; and communicate the collected illumination to a sensor comprised in the laser profiler.

19. The method of claim 18, wherein the acquiring the height of the laser profiler above the target comprises converting, via the sensor, the collected illumination to the height.

20. The method of claim 16, further comprising: comparing the determined position of the target to a predetermined target limit; and at least one of: accepting the determined position of the target in response to the determined position being within the predetermined target limit; or adjusting the target in response to the determined position of the target being outside the predetermined target limit.
